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Research article

Empowerment of accurate modeling of anaerobic membrane bioreactors by automated machine learning

Junjie Yu^{a,1}, Zhonghua Zheng^{a,1}, Jialing Ni^{b,c}, Jiayuan Ji^{d,*}

^a Department of Earth and Environmental Sciences, The University of Manchester, Manchester, M13 9PL, UK

^b Research Center for Advanced Science and Technology, The University of Tokyo, Tokyo, 153-8904, Japan

^c UTokyo LCA Center for Future Strategy, The University of Tokyo, Tokyo, 153-8904, Japan

^d Institute for Future Initiatives, The University of Tokyo, Tokyo, 113-8654, Japan

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ABSTRACT

Machine learning (ML) is a promising approach for anaerobic membrane bioreactor (AnMBR) modeling. However, its complexity poses challenges for researchers who lack expertise in ML. To address the challenges of model selection and hyperparameter optimization in ML, we employed automated machine learning (AutoML) to model the removal of chemical oxygen demand (COD) in AnMBR treating municipal wastewater. The results show that AutoML achieved better modeling performance than previously applied deep neural models, with a mean absolute percentage error of 3.11%. We further investigated the impact of expanding the feature set and increasing the amount of data on modeling performance. The results demonstrate that the features of operation time can improve performance. However, adding more data does not markedly contribute to ML-based modeling. An analysis of ensemble feature importance indicates that the concentration of COD in influent is the most important feature for predicting the removal of COD. This study demonstrates the efficacy of AutoML for modeling AnMBRs and provides insights into the modeling of wastewater treatment processes with small datasets.

1. Introduction

In the near future, humans will use and pollute freshwater at levels where water scarcity could limit food production, ecosystem functioning, and urban water supply (Jury and Vaux, 2007). The water industry has identified wastewater as a viable and sustainable source of resource recovery (Winkler and van Loosdrecht, 2022), and activated sludge is at the core of wastewater treatment technologies (van Loosdrecht and Brdjanovic, 2014). Among the wastewater treatment technologies, the anaerobic membrane bioreactor (AnMBR) is a promising solution because of its low operational cost, reduced waste sludge production, and high potential for energy recovery (Ho and Sung, 2010; Pretel et al., 2015; Robles et al., 2022). By integrating anaerobic digestion with membrane filtration, AnMBRs not only achieves effective wastewater treatment but also facilitates resource recovery, such as biogas production, rendering it an effective technology for a circular economy and sustainable development (Krzeminski et al., 2017; Moideen et al., 2023). Additionally, AnMBRs offer a fast start-up and a

smaller footprint for wastewater treatment (Robles et al., 2018). However, the development of AnMBRs primarily relies on laboratory experiments, followed by pilot-scale trial (Q. Li et al., 2024; Z. Li et al., 2024; Ren et al., 2024). Operating such systems generally consumes large amounts of chemical reagents, needs professional equipment support (Li et al., 2022), and requires months to years, rendering them time- and resource-intensive. These constraints on laboratory resources and space further limit the number and duration of reactor runs, impeding the development of AnMBRs.

To overcome these constraints, data-driven modeling, particularly machine learning (ML), has gained considerable attention as a tool for advancing AnMBR technologies and processes (Li et al., 2022; Robles et al., 2018; Wang and Li, 2024). ML models enable simulations of reactor performance by utilizing operational parameters and influent water quality as inputs and predicting effluent water quality as outputs (Li et al., 2022; Mahanna et al., 2024; Pal et al., 2024). This approach facilitates the analysis of the complex relationships between influent and effluent characteristics and offers valuable insights into reactor

* Corresponding author.
E-mail addresses: kikaen@ifi.u-tokyo.ac.jp, kikaen@tohoku.ac.jp (J. Ji).
¹ These authors contributed equally to this work.

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1. Giới thiệu

Nâng cao năng lực mô hình hóa chính xác bể phản ứng sinh học màng kỵ khí bằng học máy tự động

Bài báo nghiên cứu

Junjie Yu^{^{a,1}}, **Zhonghua Zheng**^{^{a,1}}, **Jialing Ni**^{^{b,c}},
Jiayuan Ji^{^{d,*}}

^{^{*}} Tác giả liên hệ. E-mail: kikaen@ifi.u-tokyo.ac.jp, kikaen@tohoku.ac.jp (J. Ji).
^¹ Các tác giả này đóng góp như nhau cho công trình này.

THÔNG TIN BÀI BÁO

****Từ khóa:****
- Tập dữ liệu nhỏ
- Xử lý nước thải
- Mô hình hóa dự đoán
- Độ quan trọng của đặc trưng
- Mô hình hóa dựa trên dữ liệu
- Tự động tinh chỉnh

TÓM TẮT

Nghiên cứu này nhằm mục đích tận dụng khung AutoML để mô hình hóa nhanh động học phi tuyến phức tạp của AnMBR trong quá trình xử lý nước thải. Trong khuôn khổ mô hình hóa nhanh được đề xuất, chúng tôi đã khảo sát tác động của việc tăng số lượng đặc trưng (biến dự đoán) và kích thước tập dữ liệu huấn luyện lên hiệu suất mô hình bằng cách so sánh kết quả với các mô hình hiện có dựa trên mạng nơ-ron. Ngoài ra, chúng tôi đã đề xuất một khung diễn giải học kết hợp để giải thích mô hình dựa trên AutoML. Các phát hiện của nghiên cứu này đóng góp vào việc ứng dụng rộng rãi hơn của AutoML trong mô hình hóa bể phản ứng xử lý nước thải, đồng thời cung cấp hướng dẫn giá trị để phát triển các mô hình hiệu quả cho hệ thống AnMBR.

operation and performance optimization. In addition, ML-based simulations provide a cost-effective alternative to traditional experimental methods, enabling the evaluation of numerous potential operational scenarios with minimal resource consumption. Such capabilities are particularly valuable for guiding the design and operation of AnMBR systems, particularly in contexts where real-world experiments are impractical or resource constrained.

However, the application of ML in AnMBR modeling remains challenging. One major obstacle is the significant heterogeneity of AnMBR systems, which stems from variations in the microbial community structures, physical configurations, and influent water quality. These factors fundamentally affect reactor performance and resource recovery efficiency. For instance, Ji et al. (2020) observed that the membrane pore size in an AnMBR can influence the effluent chemical oxygen demand (COD) concentrations and the reactor's microbial composition (Ji et al., 2020), thereby affecting the effluent quality and resource recovery efficiency. This variability limits the development of generalized ML models that can be reliably applied to different reactors and environmental conditions. Instead, ML models often require customization to account for the unique configurations of individual reactors. This demands for tailored modeling for flexible and adaptive approaches for ML-driven AnMBR optimization. Another complication is that developing effective ML models for AnMBR systems involves a series of complex tasks, including ML algorithm selection and hyperparameter optimization, which typically require substantial ML expertise (He et al., 2021; Karmaker ("Santu") et al., 2021). To address these challenges, automated machine learning (AutoML) has emerged as a promising solution to reduce the complexity of repetitive tasks in ML pipelines (Truong et al., 2019). Although ML models have been implemented in MBRs, their real-world uptake remains limited. AutoML has been highlighted as a promising method for facilitating the practical deployment of ML models (Lai et al., 2024). In particular, AutoML

automates key steps, such as ML algorithm selection and hyperparameter optimization, thereby reducing the technical complexity and time required for developing high-performance ML models. In addition, by democratizing access to ML techniques, AutoML has the potential to empower a wider range of users to leverage data-driven approaches in AnMBR development, thereby fostering broader adoption and innovation in this field (Li et al., 2021).

This study aims to leverage an AutoML framework for rapid modeling of the complex nonlinear dynamics of the AnMBR during wastewater treatment. Within the proposed rapid modeling framework, we investigated the impact of increasing the features (predictive variables) and training dataset size on the model performance by comparing the results with those of existing models based on neural networks. Additionally, we proposed an ensemble learning interpretation framework to explain the AutoML-based model. The findings of this study contribute to the broader application of AutoML in wastewater treatment reactor modeling and provide valuable guidance for developing effective models for AnMBR systems.

2. Methods

The overall workflow includes data collection from on-site lab-scale experiments, AutoML model development with different inputs, and ensemble ranking analysis of feature importance (Fig. 1). Data collection is described in Section 2.1. AutoML model development is detailed in Section 2.2 and 2.3, and feature importance calculations and analyses are presented in Section 2.4.

2.1. Data

The data were obtained from two AnMBRs that operated continuously in long-term experiments treating municipal sewage (Ji et al.,

Mô hình AnMBR được xây dựng như sau:

2. Phương pháp

2.1. Dữ liệu

Quy trình làm việc tổng thể bao gồm thu thập dữ liệu từ các thí nghiệm quy mô phòng thí nghiệm tại hiện trường, phát triển mô hình AutoML với các đầu vào khác nhau, và phân tích xếp hạng kết hợp về độ quan trọng của đặc trưng (Hình 1). Quy trình thu thập dữ liệu được mô tả trong Mục 2.1. Việc phát triển mô hình AutoML được trình bày chi tiết trong Mục 2.2 và 2.3, và các tính toán cùng phân tích độ quan trọng của đặc trưng được trình bày trong Mục 2.5.

****Hình 1.**** Tổng quan về quy trình làm việc. (a) Lưu đồ thí nghiệm AnMBR. (b) Khung mô hình hóa. T-R: nhiệt độ vận hành bể; pH-in: pH nước thải đầu vào; T-in: nhiệt độ nước thải đầu vào; T-env: nhiệt độ môi trường; COD-in: nhu cầu oxy hóa học của nước thải đầu vào; OD: số ngày vận hành; ORP: thế oxy hóa - khử; HRTs: thời gian lưu thủy lực; MLSS: chất rắn lơ lửng trong bùn lỏng; MLVSS: chất rắn lơ lửng bay hơi trong bùn lỏng; FLMAL: gói học máy tự động; FCN: mạng kết nối đầy đủ; CNN: mạng nơ-ron tích chập; và DenseNet: mạng tích chập kết nối dày đặc. Tree-based, permutation và SHAP biểu thị các phương pháp giải thích khác nhau cho các mô hình học máy.

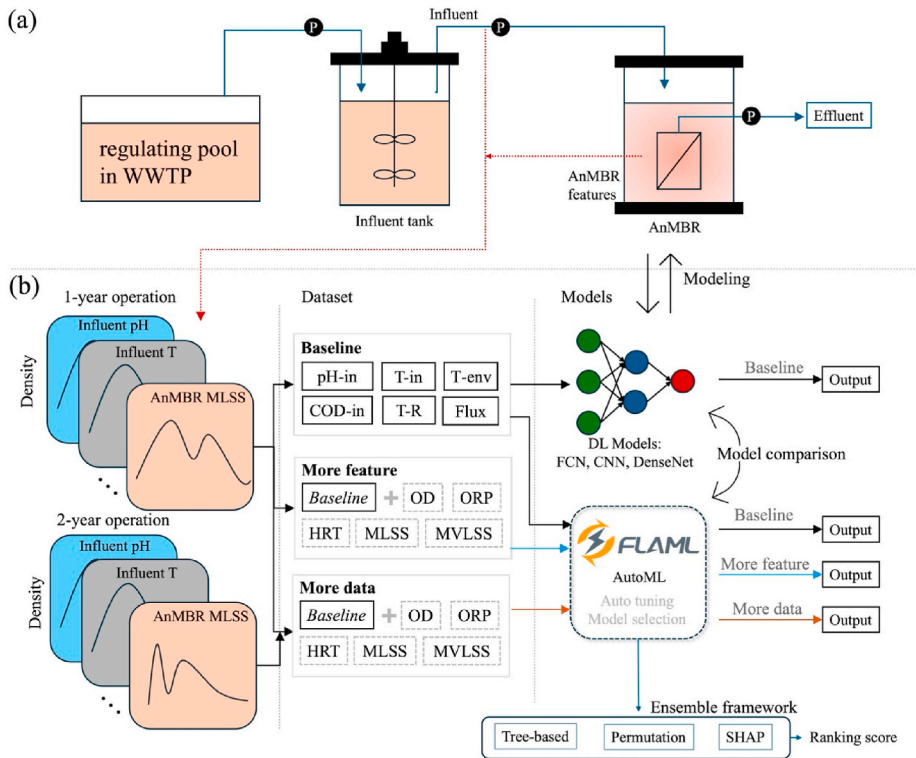


Fig. 1. Overview of the workflow. (a) Flowchart of AnMBR experiment. (b) Modeling framework. T-R: reactor operation temperature; pH-in: influent pH; T-in: influent temperature; T-env: environment temperature; COD-in: influent chemical oxygen demand; OD: operation days; ORP: oxidation-reduction potential; HRTs: hydraulic retention times; MLSS: mixed liquor suspended solids; MLVSS: mixed liquor volatile suspended solids; FLMAL: the automated machine learning package; FCN: fully connected network; CNN: convolutional neural network; and DenseNet: densely connected convolutional network. Tree-based, permutation, and SHAP indicate different interpretation methods for machine learning models.

2.2. Mô hình hóa

2020, 2021a, 2021b). The volume of each reactor was 20 L, and the reactors were installed at a wastewater treatment plant (WWTP) for better access to raw municipal sewage. The membrane modules used in both AnMBRs were hollow fiber polyvinylidene difluoride membranes, but with pore sizes of $0.4 \mu\text{m}$ and $0.05 \mu\text{m}$. The operating temperature was maintained between 20°C and 25°C using water baths (NTT-20S, EYELA, Japan). Hydraulic retention times (HRTs) of 4–24 h were applied to AnMBR1, whereas HRTs of 10–24 h were applied to AnMBR2. Additional details, including the basic water quality of the raw municipal wastewater, experimental equipment, and conditions, have been reported in previous studies (Ji et al., 2020, 2021a, 2021b). The COD, mixed liquor suspended solids (MLSS), and mixed liquor volatile suspended solids (MLVSS) were measured by standard methods (Bridgewater et al., 2017). The MLSS and MLVSS data were sampled weekly, and these data were filled using linear interpolation to match the frequency of COD data. The oxidation-reduction potential (ORP) of the anaerobically activated sludge was measured using an ORP meter (TOADKK, RM-30P, Japan). A total of 11 features were included in this study, including the reactor operation days (OD; Unit: day), reactor operation temperature (T-R; Unit: $^\circ\text{C}$), influent temperature (T-in; Unit: $^\circ\text{C}$), environment temperature (T-env; Unit: $^\circ\text{C}$), influent pH (pH-in; Unitless), influent COD (COD-in; Unit: mg/L), flux (Unit: m/day), ORP (Unit: mV), HRT (Unit: hour), MLSS (Unit: mg/L), and MLVSS (Unit: mg/L). The output was the COD removal rate (COD-re; Unit: %). The total sample size of this dataset was 185, which was divided into training and testing sets using a 9:1 split ratio (166:19) in previous study (Li et al., 2022). The above data was obtained by running the reactors in 50–414 days.

We designed comparative studies to investigate: (1) the permanence of AutoML, (2) the impact of feature selection on modeling performance, and (3) the effects of data volume (Fig. 1). First, for comparing with the previous models, we used the same training set and testing data as in a previous study that applied deep learning models, in which T-R, T-in, T-env, pH-in, COD-in, and flux were used as inputs and COD-re was used as a label (Li et al., 2022). The results of deep learning networks from Li et al. (2022), including a fully connected network (FCN), a convolutional neural network (CNN), and a densely connected convolutional network (DenseNet), were adopted. These existing models serve as baselines (Fig. 1). Subsequently, we examined feature contribution by integrating OD, ORP, HRT, MLSS, and MLVSS (more feature modeling experiment in Fig. 1). These variables were analysed through both individual and combined inclusion to delineate their effects on modeling performance. Lastly, to explore whether the size of the training dataset would improve ML model performance, we used additional data collected from longer operational times of the same AnMBRs (more data modeling experiment in Fig. 1). The sample size of the additional data was 135, which was obtained by running the reactors in 7–546 days. In addition to benchmarking against Li et al. (2022), we employed 10-fold cross-validation to evaluate the AutoML models. During 10-fold cross-validation, 10 independent models were trained, with each iteration using a distinct 90% training and 10% testing split. This approach assesses the variance in model performance and ensures that the reported results are robust to data perturbations.

2.2. Modeling

The model of AnMBR is constructed as follows:

$$Y = f_\theta(X), \quad (1)$$

where Y is the COD-re, X is the matrix of input features, and f_θ is the model parameterized with θ . Data-driven modeling aims to estimate the optimal set of parameters, θ , that best represents the given data.

However, identifying suitable model parameters can be difficult because the process often depends on appropriately configuring hyperparameters (e.g., learning rate, batch size, epochs for neural networks,

estimator number, learning rate, and maximum depth of tree for tree-based models) to guide the learning process. Hyperparameters are settings that influence the training process and model performance but are not learned through the training process.

Hyperparameter tuning is a crucial step in optimizing the model's performance. Common methods for hyperparameter optimization include grid search, random search, Bayesian optimization, and genetic algorithms. Although these methods assist in hyperparameter tuning, they often require substantial ML expertise and computational resources, which limit their accessibility. Consequently, AutoML has been developed to streamline the process of hyperparameter tuning and make it more accessible.

In this study, we employed a Fast Library for AutoML and tuning (FLAML), supporting fast and economical automatic tuning for machine learning (Wang et al., 2021b). This library selects a search order optimized for both computational cost and model error, iteratively selecting the models, hyperparameters, sample size, and resampling strategy. FLAML demonstrated remarkable performance, outperforming other leading libraries of AutoML (e.g., Auto-sklearn, H2O AutoML, and Alpine Meadow) (Wang et al., 2021b). Studies from environmental sciences, atmospheric sciences, and climate sciences are benefited from this package (Dong et al., 2025; Xia et al., 2023; Zheng et al., 2023). As tree-based models are suitable for tabular data, they were selected as base candidates for the FLAML model selection process, including the random forests (RF) (Breiman, 2001), extremely randomized trees (Geurts et al., 2006), XGBoost (a scalable end-to-end tree boosting system) (Chen and Guestrin, 2016), LightBGM (a highly efficient gradient boosting decision tree) (Ke et al., 2017), and Catboost (gradient boosting with categorical features support) (Dorogush et al., 2018). FLAML implements a Cost-Frugal Optimization (CFO) strategy, which starts with a low-cost configuration (e.g., small model size) and iteratively moves to more complex configurations only when the performance gain justifies the additional computational cost (Wu et al., 2021). Thus, this method can control cost during the optimization process compared to traditional hyperparameters optimization processes. Further, BlendSearch was applied by default, which is a hierarchical search algorithm that combines global exploration with local optimization to minimize the total cost spent in finding good configurations (Wang et al., 2021a). The optimization process was constrained by a global time budget of 300 s. The optimization will be terminated when time budget is exhausted. To prevent overfitting, we adopted a 5-fold cross-validation strategy with a mean squared error (MSE) criterion during each AutoML optimization. Notably, the search spaces for the select learners above are dynamically included the number of estimators, number of leaves, learning rate, and regularization parameters. The specific search ranges were then determined by CFO strategy based on the training dataset size. The optimal sets of hyperparameters and settings of all models are provided in Supporting Information.

2.3. Evaluation

The evaluation metrics utilized in this study included the MSE, mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2), which were calculated as follows:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (2)$$

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (3)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| * 100, \quad (4)$$

$$f_{\theta}(X), \quad (1)$$

trong đó Y là COD-re, X là ma trận các đặc trưng đầu vào, và f_{θ} là mô hình được tham số hóa với θ . Mô hình hóa dựa trên dữ liệu nhằm ước tính tập tham số tối ưu θ đại diện tốt nhất cho dữ liệu đã cho.

Nghiên cứu này sử dụng các chỉ số đánh giá gồm MSE, sai số tuyệt đối trung bình (MAE), sai số căn quân phương (RMSE), sai số phần trăm tuyệt đối trung bình (MAPE) và hệ số xác định (R^2). Các chỉ số được tính như sau:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (2)$$

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (3)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| * 100, \quad (4)$$

2.4. Nghiên cứu loại trừ

2.3. Đánh giá

2.5. Điểm xếp hạng đặc trưng

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (5)$$

where y_i and $y^{\wedge}_i$ are the observations and model predictions, respectively, and n is the number of samples. MAE is the average of the absolute differences between y_i and $y^{\wedge}_i$. RMSE is the square root of the mean of the squared differences between y_i and $y^{\wedge}_i$. It assigns higher weights to larger errors owing to the squaring of errors compared with MAE. RMSE and MAE represent the precision of model predictions. MAPE is the average percentage difference between y_i and $y^{\wedge}_i$. An R^2 value of one signifies a perfect fit of the model to the data. When R^2 value is close to zero, it indicates that the model has minimal explanatory power over the target variable. When R^2 value is negative, it indicates that prediction error exceeds the inherent variance of the dataset. In general, the R^2 value is better when closer to 1. Notably, the units of MAE and RMSE in this study are percentages (%) (same as the output unit), and the unit of MAPE is also percentage (%) (relative error that calculated by equation (4)).

While error metrics (e.g., RMSE) assess the overall predictive performance, they do not explicitly quantify the agreement between the predicted values and the ground truth (Martin Bland and Altman, 1986). Therefore, to evaluate the concordance between the predicted COD removal efficiency and the ground truth, the Bland-Altman analysis was performed (Martin Bland and Altman, 1986). The Bland-Altman method analyses the difference between the two methods against their average. In this study, the difference is calculated as the ground truth subtracting the prediction. A non-zero difference reveals that the model generally overestimates (difference <0) or underestimates (difference >0) the measurement. If the limits of agreement did not exceed the professionally acceptable value range and 95% of the differences lie within 95% limits of agreement, then the consistency of the two methods was considered to be good (Li et al., 2022). The 95% limits of agreement were determined by adding and subtracting 1.96 standard deviations to the mean difference.

2.4. Ablation studies

We conducted ablation studies to evaluate the contributions of different features to the performances of the ML models. An ablation study is a technique used to assess the impact of specific components (e.g., features, parameters, or data) by removing or modifying them and observing the resulting changes in model performance. Specifically, we assessed the impact of OD, ORP, HRT, MLSS, and MLVSS on model performance. By comparing the performance of models with and without specific features, we can infer how these features influence modeling. Additionally, given that we collected more data than the previous study, we investigated the effect of this extended dataset on modeling performance. Throughout the ablation studies, the training and testing data, as well as the model training parameters, were maintained consistently with the previous setup.

2.5. Feature ranking score

We developed an ensemble framework using ranking score tailored for AutoML tasks. This framework combines tree-based feature importance, permutation feature importance, and shapley additive explanations (SHAP) values (Schreck et al., 2023; Yu et al., 2025; Zheng et al., 2023), and produces an ensemble score with feature importance ranking. This approach provides a more balanced assessment by combining multiple perspectives, rather than relying on a single method.

In tree-based models, feature importance is derived directly from the structure of the models. For example, in RF, importance is calculated

using the Gini importance, which measures the normalized reduction in the Gini criterion contributed by each feature. In this paper, this method is referred to as a tree-based method.

The permutation method calculates feature importance by evaluating the change in model performance when the values of a feature are randomly shuffled (Breiman, 2001), which can be applied to any model using tabular data. Specifically, the importance of feature j is defined as

$$i_j = s - \frac{1}{K} \sum_{k=1}^K s_{k,j}, \quad (6)$$

where i_j represents the importance of feature j , s represents the reference model score (RMSE), and $s_{k,j}$ represents the model score after shuffling feature j at k^{th} repetition. K is the total number of repetitions (30).

In addition, we employed SHAP, a model-agnostic framework for explaining predictions based on Shapley values (Lundberg et al., 2018; Lundberg and Lee, 2017). The SHAP method calculates the contribution of each feature to the output of the model (SHAP value) as follows:

$$\phi_i(f, x) = \sum_{S \subseteq S_{\text{all}} \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [f_x(S \cup \{i\}) - f_x(S)], \quad (7)$$

where $\phi_i(f, x)$ represents the SHAP value for feature i , $f(S)$ represents the model output for a subset of features S ($S \subseteq (0, 1)^M$), and M represents the set of all M input variables. $|S|$ is the number of nonzero entries in S , and the summation iterates over all possible feature subsets. The contribution of feature i is determined by comparing the model output with and without this feature, weighted by the subset size.

The feature importance metrics obtained from the aforementioned methods are difficult to compare because of their different underlying principles. To address this, we introduced a “ranking score” metric to standardize the comparison of feature importance across the methods (Zheng et al., 2023). Feature importance values obtained from each method were sorted in ascending order, and each feature received a ranking corresponding to its position in the order-list. Specifically, the least important feature has a lowest score (1), and the second least important is 2, with higher scores given to increasingly important features. This process ensured that the ranking scores ranged from one (indicating the least important feature) to the total number of features (indicating the most important feature). By normalizing the feature importance values in this manner, the feature importance from the different methods was consistent, facilitating direct comparison.

3. Results and discussion

3.1. AutoML for efficient AnMBR modeling

We constructed the AutoML model on the same dataset (including features and targets) as Li et al. (2022) and benchmarked its performance against their results. The results showed that AutoML achieves competitive performance compared to the reported results of previous deep learning models (FCN, CNN, and DenseNet) (Table 1 and Fig. 2). The negative R^2 values observed for previously deep learning architectures may originate from overfitting, limited dataset size, lack of relevant features, and model suitability for the task (Jiang et al., 2025). By using AutoML, the R^2 value increased from negative to positive, highlighting a marked improvement in model performance. Further, we evaluated the AutoML using 10-fold cross-validation. The narrow RMSE confidence interval ($\pm 0.33\%$) validates the robustness of AutoML. Notably, tree-based models were used as base learners for the AutoML tasks. These models exhibit notable differences from neural networks, particularly in handling tabular data (Grinsztajn et al., 2022; Shwartz-Ziv and Armon, 2022). Existing evidence also suggests that tree-based algorithms often outperform deep learning methods when applied to structured tabular data and do not require extensive data preprocessing (McElfresh et al., 2023). Consequently, the adoption of

$$s_i = s - \frac{1}{K} \sum_{k=1}^K s_{k,i}, \quad (6)$$

Ngoài ra, chúng tôi sử dụng khung làm việc SHAP (Lundberg et al., 2018; Lundberg and Lee, 2017). Khung độc lập với mô hình này giải thích các dự đoán dựa trên giá trị Shapley. Phương pháp SHAP tính toán mức đóng góp của từng đặc trưng vào đầu ra mô hình (giá trị SHAP) như sau:

$$\phi_i(f, x) = \sum_{S \subseteq S_{\text{all}} \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [f_x(S \cup \{i\}) - f_x(S)], \quad (7)$$

****Bảng 1.**** Hiệu suất mô hình và so sánh.

3.3. Tác động của khối lượng dữ liệu đến hiệu suất dự đoán

Các chỉ số độ quan trọng của đặc trưng từ những phương pháp nêu trên rất khó so sánh do khác biệt về nguyên lý nền tảng. Để giải quyết vấn đề này, chúng tôi đưa vào chỉ số “điểm xếp hạng” (Zheng et al., 2023). Chỉ số này chuẩn hóa việc so sánh độ quan trọng của đặc trưng giữa các phương pháp. Các giá trị độ quan trọng của đặc trưng từ mỗi phương pháp được sắp xếp theo thứ tự tăng dần. Mỗi đặc trưng nhận một thứ hạng tương ứng với vị trí trong danh sách sắp xếp. Cụ thể, đặc trưng ít quan trọng nhất nhận điểm thấp nhất (1), và đặc trưng ít quan trọng thứ hai nhận điểm 2. Điểm số cao hơn được gán cho các đặc trưng có độ quan trọng tăng dần. Quy trình này đảm bảo điểm xếp hạng nằm trong dải từ 1 (đặc trưng ít quan trọng nhất) đến tổng số đặc trưng (đặc trưng quan trọng nhất). Bằng cách chuẩn hóa này, độ quan trọng của đặc trưng giữa các phương pháp đạt tính nhất quán, giúp việc so sánh trực tiếp trở nên thuận lợi.

****Hình 4.**** Hiệu suất mô hình với các đặc trưng đầu vào khác nhau. Baseline biểu thị việc sử dụng T-R, T-in, T-env, pH-in, COD-in và thông lượng (flux) làm đặc trưng. Thanh sai số biểu thị khoảng tin cậy 95% của các chỉ số sai số từ kiểm định chéo 10 lần.

3. Kết quả và thảo luận

3.1. AutoML cho mô hình hóa AnMBR hiệu quả

Table 1

Model performance and comparison.

	RMSE (%) ^a	MAE (%) ^a	MAPE (%) ^b	R ²	Evaluation Methods
AutoML	2.47 [2.14, 2.81]	1.90 [1.68, 2.11]	2.19 [1.93, 2.46]	0.44 [0.22, 0.65]	10-fold CV ^c
AutoML	3.09	2.76	3.11	0.47	Li et al. (2022) ^d
FCN	6.29	5.71	6.49	−1.19	Li et al. (2022) ^d
CNN	5.98	5.34	6.02	−0.97	Li et al. (2022) ^d
Dense	4.69	4.34	4.93	−0.21	Li et al. (2022) ^d

^a The unit is equal to COD-re (%).

^b The unit is percentage according to equation (4).

^c The error metrics are reported as the mean and 95% confidence intervals of the error metrics from 10-fold cross validation (CV) using Student's t-based distribution.

^d The model was evaluated by splitting the data using the method from Li et al. (2022).

tree-based AutoML not only enhances the model accuracy but also improves the efficiency of the AnMBR modeling workflow.

Furthermore, Bland-Altman analysis was employed to evaluate the consistency between predictions and observations (the ground truth), providing insights into the predictive accuracy and stability of the models. It shows the average difference (bias) and calculates the limits of agreement, which capture the expected range for most differences. A wider limit of agreement boundary indicates poorer consistency between the model and the observations. The AutoML results were distributed in a lower limit of agreement range than those of the deep learning models, with a more uniform error distribution (Fig. 3). This suggests that the AutoML approach yields more consistent and stable predictions. In our analysis, the majority of data points were within the 95% confidence interval, demonstrating a strong agreement between the observed values and model predictions.

For the deep learning models, a trend was observed in which the difference between the predicted and observed values decreased as the mean value increased. By contrast, the AutoML results displayed no discernible pattern, with differences evenly distributed across the range of mean values, highlighting the stability and robustness of AutoML. It is

noteworthy that all the analysis results show a positive mean difference (Fig. 3). This indicates a systematic tendency to underestimate the COD removal rate using these models. This underestimation may be attributed to the imbalanced data distribution between the training and testing sets. Additionally, data-driven models tend to smooth out the noise introduced by measurement uncertainty and the dynamic nature of the biological process. Despite this, the AutoML achieved an RMSE of 3.09(%), which is lower than the standard deviation of the experimental COD removal efficiency reported in long-term AnMBR operations (Robles et al., 2022; Uman et al., 2021). This result indicates that the model's overall accuracy is sufficient to describe the AnMBR dynamics. However, as we discussed, overfitting, limited dataset size, model suitability for the task, and testing methods affect the performance, thus the comparison results are indicative rather than definitive.

3.2. Improved ML performance with operation time as a feature

The treatment efficiency of the AnMBR is not only affected by T-R, T-in, T-env, pH-in, COD-in, and flux but also by the reactor operation conditions. For instance, the microbial community structure and activity in an AnMBR are key factors that affect its efficiency (Hu et al., 2017; Xie et al., 2014). However, measuring microbial community structure is a difficult task because it is time-consuming and costly, which restricts the availability of community structure data. Consequently, this limitation may also hinder data-driven AnMBR modeling from effectively uncovering the underlying mechanisms.

Based on the hypothesis that additional information may correlate with operation time, we incorporated reactor OD into the modeling process, assuming that OD may be a potential proxy for the microbial dynamics. The inclusion of OD resulted in a substantial improvement in model performance compared to the baseline feature set (T-R, T-in, T-env, pH-in, COD-in, and flux) (Fig. 4 and Fig. S1), with the mean of R² value increasing from 0.44 to 0.55 and mean of RMSE decreasing from 2.47 to 2.20 (Fig. 4). Spearman correlation analysis indicates a weak association between OD and the target variable, suggesting a low likelihood of data leakage (see supplementary material). Thus, such enhancement suggests that the additional information carried by OD may have been effectively learned by the ML model. It is important to note that rather than being a direct causal parameter, OD acts as a surrogate variable capturing unmeasured temporal dynamics inherent to the AnMBRs process. This includes the gradual acclimatization of the microbial community (e.g., biofilm development) and the cumulative

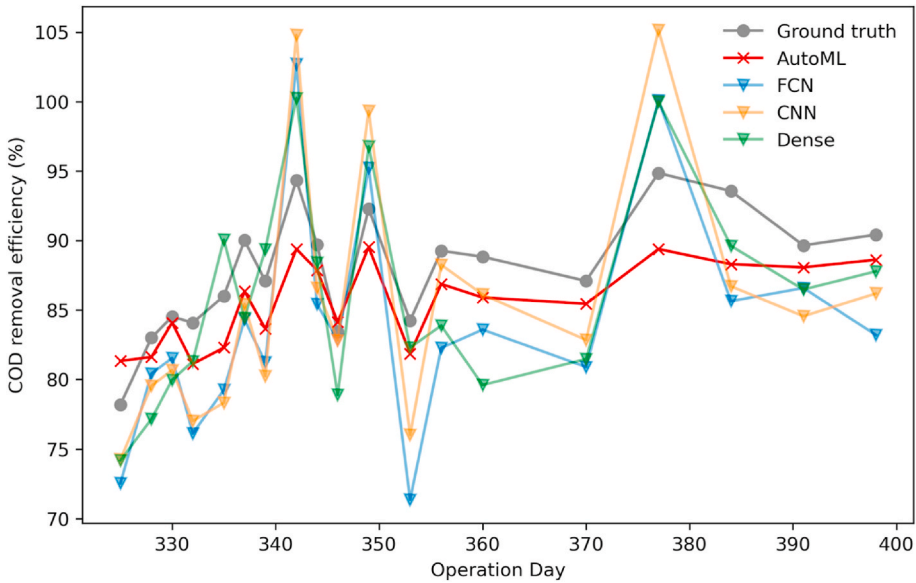


Fig. 2. Model prediction results and comparison with actual measured values.

3.5. Hạn chế và triển vọng tương lai

4. Kết luận

Tính khả dụng của dữ liệu
Các tác giả không có quyền chia sẻ dữ liệu.

Tài liệu tham khảo

Breiman, L., 2001. Random forests. Mach. Learn. 45, 5–32. <https://doi.org/10.1023/A:1010933404324>.

Chen, T., Guestrin, C., 2016. XGBoost: a scalable tree boosting system. Trong: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '16. Association for Computing Machinery, New York, NY, USA, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>.

Dong, W., Yuan, H., Lin, W., Liu, Z., Xiang, J., Wei, Z., Li, L., Li, Q., Dai, Y., 2025. A global urban tree leaf area index dataset for urban climate modeling. Sci. Data 12, 426. <https://doi.org/10.1038/s41597-025-04729-y>.

****Hình 5.**** Phân tích Bland-Altman của các mô hình AutoML. (a) Baseline, (b) thêm OD, (c) thêm OD, ORP, HRT, MLSS và MLVSS, (d) thêm tất cả trừ OD vào đặc trưng. Chênh lệch của COD-re được tính bằng giá trị thực tế trừ đi giá trị dự đoán. Đường trung tâm thể hiện chênh lệch trung bình của COD-re. Mỗi đồ thị có ba đường nằm ngang từ trên xuống dưới. Đường trên cùng là giới hạn đồng thuận trên (chênh lệch trung bình + 1.96 × độ lệch chuẩn). Đường giữa biểu thị chênh lệch trung bình. Đường dưới cùng là giới hạn đồng thuận dưới (chênh lệch trung bình - 1.96 × độ lệch chuẩn).

Kết quả nhấn mạnh yêu cầu kiểm tra kỹ nguồn, chất lượng và độ phù hợp khi áp dụng ML cho tập dữ liệu nhỏ. Tăng lượng dữ liệu mà không xét tính dị thể hay độ liên quan có thể làm giảm độ chính xác và tính vững. Các nghiên cứu tương lai nên tập trung vào phương pháp tích hợp hiệu quả dữ liệu từ nhiều nguồn. Nghiên cứu cần giảm thiểu các độ lệch tiềm ẩn để đảm bảo độ tin cậy và độ chính xác của mô hình ML.

Chúng tôi sử dụng phương pháp điểm xếp hạng (Mục 2.5) để đánh giá độ quan trọng của đặc trưng. Kết quả từ ba phương pháp chỉ ra rằng COD-in là đặc trưng quan trọng nhất trong mọi mô hình (Hình 7). Đối với mô hình không có OD và được huấn luyện không dùng dữ liệu bổ sung, T-env là đặc trưng quan trọng thứ hai (Hình 7(a)). Ngược lại, với mô hình không có OD nhưng được huấn luyện cùng dữ liệu bổ sung, pH-in trở thành đặc trưng quan trọng thứ hai (Hình 7(e)). Khi bổ sung OD, biến này trở thành đặc trưng quan trọng thứ hai, nếu bật vai trò quan trọng của nó trong quá trình mô hình hóa (Hình 7(c–g)). Nhìn chung, việc COD-in là đặc trưng quan trọng nhất để dự đoán hiệu quả loại bỏ COD của AnMBR rất hợp lý. Thông số này đóng vai trò là chỉ báo quan trọng về tải trọng của bể phản ứng. Ngược lại, các đặc trưng khác như nhiệt độ môi trường, nhiệt độ nước thải đầu vào và pH có độ quan trọng thấp hơn trong mô hình. Điều này cho thấy các thông số môi trường trên có tác động hạn chế đến hiệu suất của AnMBR.

Sự biến thiên về độ quan trọng của đặc trưng phản ánh độ bất định liên quan đến các phương pháp toán học khác nhau dùng để tính toán độ quan trọng của đặc trưng. Tuy nhiên, bất chấp những dao động này, các đặc trưng hàng đầu (COD-in và/hoặc OD) luôn xếp thứ hạng cao nhất trong mọi phương pháp và mô hình. Điều này cho thấy dù các biến xếp hạng thấp hơn tiềm ẩn độ bất định, việc xác định các đặc trưng chính vẫn có tính vững chắc. Ưu thế vượt trội của COD-in nếu bật mức độ nhạy cảm đối với tải trọng COD. Điều này cho thấy COD-in có thể đóng vai trò như tín hiệu cảnh báo để dự đoán COD-re. Ví dụ, khi phát hiện tải trọng COD cao, hệ thống có thể tự động giảm lưu lượng dòng vào. Việc này giúp quá trình sinh học có thêm thời gian xử lý nước. Độ quan trọng của OD nhấn mạnh bản chất không dừng của quá trình AnMBR (ví dụ: bám bẩn màng và sự trưởng thành của sinh khối). Điều này ngụ ý việc giám sát quá trình không nên chỉ dựa vào các số đọc cảm biến tức thời. Quy trình giám sát cũng phải kết hợp các kế hoạch bảo trì phụ thuộc vào thời gian.

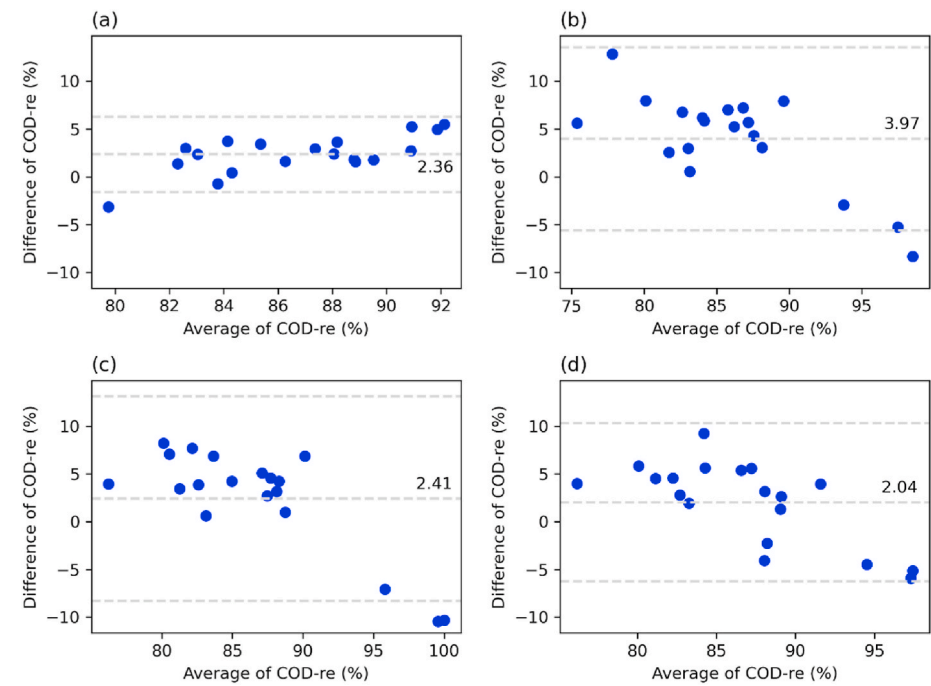


Fig. 3. Bland-Altman analysis of (a) AutoML, (b) FCN, (c) CNN, and (d) Dense. The difference of COD-re is calculated by ground truth substrates prediction. The values of the central line are the mean difference of COD-re. The three horizontal lines in each plot, from top to bottom, represent the upper limit of agreement (mean difference + 1.96 × standard deviation), the mean difference, and the lower limit of agreement (mean difference - 1.96 × standard deviation). The results were obtained using the same dataset from [Li et al. \(2022\)](#).

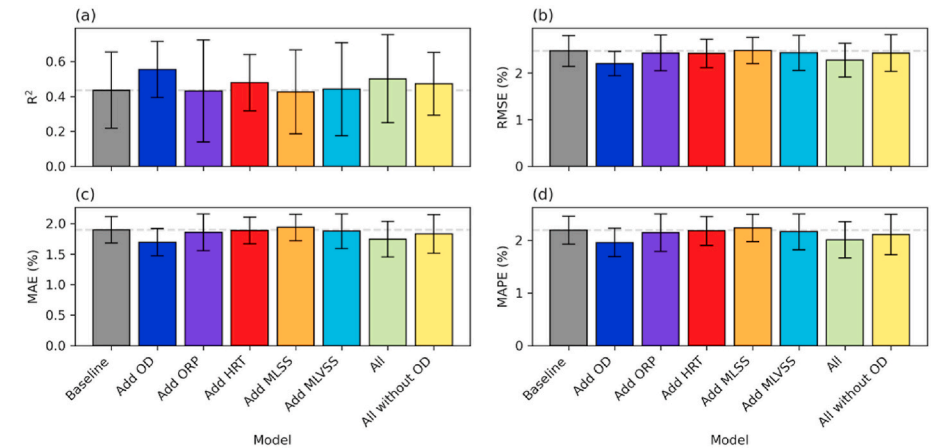


Fig. 4. Model performance with different input features. Baseline indicates using T-R, T-in, T-env, pH-in, COD-in, and flux as features. Error bar is the 95% confidence intervals of the error metrics from 10-fold cross validation.

changes in membrane surface properties over time. By incorporating this temporal dimension, the model effectively learns such non-stationary behaviors of the reactor during long-term operation. Furthermore, the inclusion of the OD may pose a challenge for extrapolation. Despite the good performance observed in this study, tree-based models often struggle to make reliable predictions beyond the training data range.

The inclusion of ORP, HRT, MLSS, and MLVSS data did not markedly contribute to the model performance. When OD, ORP, HRT, MLSS, and MLVSS were used simultaneously to train the model, the overall effect was similar to that of the inclusion of OD alone. The Bland-Altman analysis showed that incorporating OD into the model reduced the differences between the prediction and ground truth, rendering the difference more evenly distributed around zero compared to models without operation time ([Fig. 5](#) and [Fig. S2](#)). This further confirms that including the operation time improves the modeling performance,

whereas adding other input parameters yields only a limited improvement. The limited contribution of typically parameters (e.g., ORP, HRT, MLSS, and MLVSS) likely reflects the specific constraints of the data to this study, such as measurement frequency and signal-to-noise ratio, rather than a lack of physical and biological relevance. Additionally, feature collinearity may have masked their importance, as the model could derive sufficient information from other correlated predictors. These results highlight the importance of feature selection for modeling.

3.3. Impact of data volume on predictive performance

ML-based modeling is a data-driven approach whose performance primarily depends on the quality and quantity of data ([Fan and Shi, 2022](#); [Liu et al., 2023](#)). However, when applied to experimental datasets, ML has inherent challenges in terms of data availability and reliability.

Hu, D., Su, H., Chen, Z., Cui, Y., Ran, C., Xu, J., Xiao, T., Li, X., Wang, H., Tian, Y., Ren, N., 2017. Performance evaluation and microbial community dynamics in a novel AnMBR for treating antibiotic solvent wastewater. *Bioresour. Technol.* 243, 218–227. <https://doi.org/10.1016/j.biortech.2017.06.095>.

Ji, J., Chen, Y., Hu, Y., Ohtsu, A., Ni, J., Li, Y., Sakuma, S., Hojo, T., Chen, R., Li, Y.-Y., 2021a. One-year operation of a 20-L submerged anaerobic membrane bioreactor for real domestic wastewater treatment at room temperature: pursuing the optimal HRT and sustainable flux. *Sci. Total Environ.* 775, 145799. <https://doi.org/10.1016/j.scitotenv.2021.145799>.

Faraway, J.J., Augustin, N.H., 2018. When small data beats big data. *Role Stat. Era Big Data* 136, 142–145. <https://doi.org/10.1016/j.spl.2018.02.031>.

Grinsztajn, L., Oyallon, E., Varoquaux, G., 2022. Why do tree-based models still outperform deep learning on typical tabular data? Trong: Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., Oh, A. (Chủ biên), *Advances in Neural Information Processing Systems*. Curran Associates, Inc., pp. 507–520.

Geurts, P., Ernst, D., Wehenkel, L., 2006. Extremely randomized trees. *Mach. Learn* 63, 3–42. <https://doi.org/10.1007/s10994-006-6226-1>.

Ho, J., Sung, S., 2010. Methanogenic activities in anaerobic membrane bioreactors (AnMBR) treating synthetic municipal wastewater. *Bioresour. Technol.* 101, 2191–2196. <https://doi.org/10.1016/j.biortech.2009.11.042>.

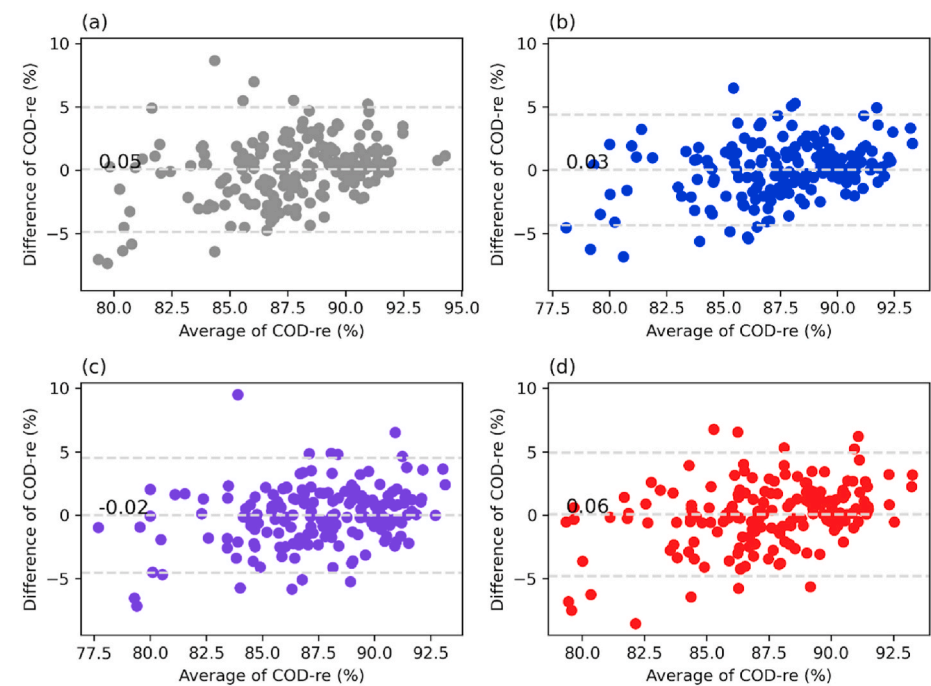


Fig. 5. Bland-Altman analysis of AutoML models. (a) Baseline, (b) add OD, (c) add OD, ORP, HRT, MLSS, and MLVSS, and (d) add all without OD in the feature. The difference of COD-re is calculated by ground truth minus prediction. The values of the central line are the mean difference of COD-re. The three horizontal lines in each plot, from top to bottom, represent the upper limit of agreement (mean difference + 1.96 × standard deviation), the mean difference, and the lower limit of agreement (mean difference - 1.96 × standard deviation).

Laboratory experiments often yield small sample sizes because of constraints such as time, cost, and space availability. Moreover, the experimental data are susceptible to various sources of error, including human and instrumental errors. These limitations increase the difficulty of applying ML-based modeling approaches effectively. Furthermore, the ability of a small, high-quality dataset to yield better inferences than a larger, low-quality one is often debatable (Faraway and Augustin, 2018; Kokol et al., 2022). To explore this, we examined the impact of increasing the amount of experimental data on AnMBR modeling. We incorporated additional data and compared the results using the same testing sets to ensure uniform evaluation. The additional data originated from the same AnMBR reactors, but with a different operational time range. We utilized the original dataset (M1, 50–414 d) and the extended dataset (M2, 7–546 d) to establish three validation schemes based on

10-fold cross-validation (Fig. 6). First, M1T1 represents the baseline performance using standard cross-validation on M1, in which T1 indicates the testing data were split from M1. Second, for M2T1, the additional data was integrated into the training folds only, allowing the model to learn from a broader range (M2) while being tested on the T1. Finally, M2T2 involves standard cross-validation on the full M2 dataset, in which T2 was split from M2.

A comparison between the M1T1 and M2T1 groups indicated that adding more data decreased the R^2 value and increased RMSE, MAE and MAPE (Fig. 6). A comparison between M1T1 and M2T2 also showed that adding more data increased the prediction error, with M2T2 showing worse modeling performance (Fig. 6). These findings suggest that the addition of more data is not necessarily beneficial for modeling. A likely explanation is that data collected under varying experimental conditions

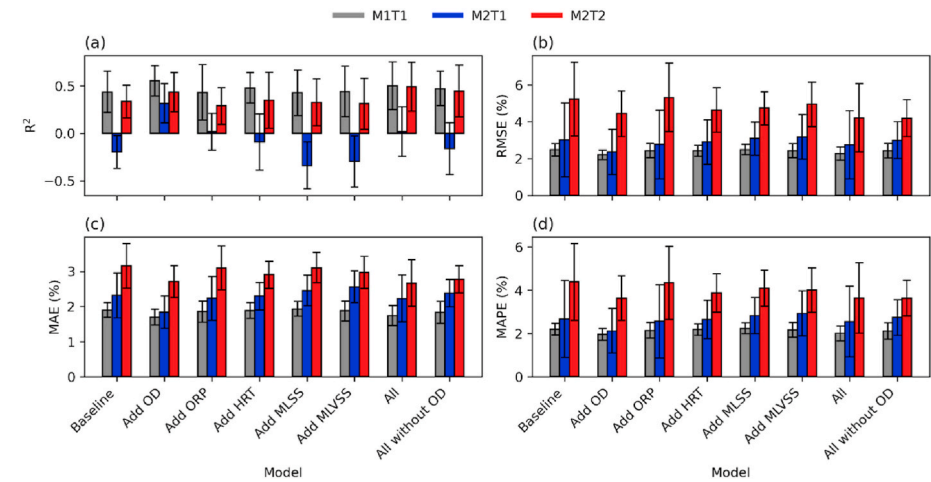


Fig. 6. Impact of dataset size on model performance. M1: original dataset with 50–414 operation days. M2: extended dataset with 7–546 operation days. M1T1: the baseline performance using standard cross-validation on M1. M2T1: the additional data was integrated into the training folds only. M2T2: standard cross-validation on the full M2 dataset.

Karmaker ("Santu"), S.K., Hassan, Md.M., Smith, M.J., Xu, L., Zhai, C., Veeramachaneni, K., 2021. AutoML to date and beyond: challenges and opportunities. *ACM Comput. Surv.* 54. <https://doi.org/10.1145/3470918>.

Ji, J., Sakuma, S., Ni, J., Chen, Y., Hu, Y., Ohtsu, A., Chen, R., Cheng, H., Qin, Y., Hojo, T., Kubota, K., Li, Y.-Y., 2020. Application of two anaerobic membrane bioreactors with different pore size membranes for municipal wastewater treatment. *Sci. Total Environ.* 745, 140903. <https://doi.org/10.1016/j.scitotenv.2020.140903>.

Jiang, F., Zheng, Z., Coe, H., Healy, R.M., Poulain, L., Gros, V., Zhang, H., Li, W., Liu, D., West, M., Topping, D., Riemer, N., 2025. Integrating simulations and observations: a foundation model for estimating the aerosol mixing state index. *ACS EST Air* 2, 877–890. <https://doi.org/10.1021/acsestair.4c00329>.

Jury, W.A., Vaux, H.J., 2007. The emerging global water crisis: managing scarcity and conflict between water users. *Tronq: Advances in Agronomy*. Elsevier, pp. 1–76. [https://doi.org/10.1016/S0065-2113\(07\)95001-4](https://doi.org/10.1016/S0065-2113(07)95001-4).

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y., 2017. LightGBM: a highly efficient gradient boosting decision tree. *Tronq: Advances in Neural Information Processing Systems*. Curran Associates, Inc.

introduce variability, biases, and specific errors, which collectively degrade the model performance. Spearman's correlation analysis of M1 and M2 further revealed that the features in M2 were less correlated with COD removal than those in M1 (Figure S3 and Figure S4). In other words, the relevance of data is important for modeling. The irrelevance of data may stem from distribution shift in influent/AnMBR states or unmeasured disturbances, which can introduce noise that outweighs the benefits of a larger sample size. Therefore, the degrading performance with larger datasets should be interpreted as a signal of including the irrelevant data. This result highlights the need for data curation that prioritize high-quality and context-specific data rather than broad but indiscriminate data aggregation.

These results highlight the importance of carefully considering the data sources, quality, and applicability when applying ML to small experimental datasets. Increasing the volume of data without accounting for heterogeneity or relevance to the modeling can lead to diminished model accuracy and robustness. Future studies should focus on methods to effectively integrate multi-source data while mitigating potential biases, ensuring the reliability and precision of ML models for experimental applications.

3.4. Ensemble feature-based importance analysis

Using the ranking score approach (Section 2.4) to evaluate feature importance, the results of the three methods indicated that COD-in was the most important feature of all models (Fig. 7). For the model without

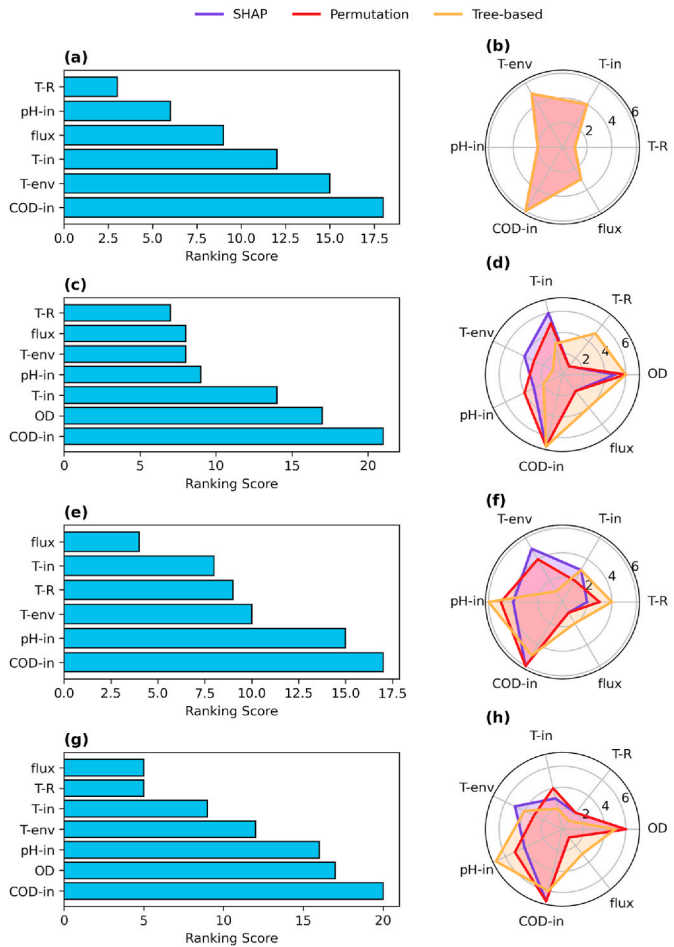


Fig. 7. Ranking score of models. (a) model trained with original data, (c) model trained with original data and additional data, (e) same as (a) with OD added in the feature, and (g) same as (c) with OD in the feature. (b), (d), (f), and (h) are the components of the ranking score of (a), (c), (e), and (g), respectively.

OD and trained without additional data, T-env appeared to be the second most important feature (Fig. 7 (a)), whereas for the model without OD but trained with additional data, pH-in became the second most important feature (Fig. 7 (e)). When the OD was included, it became the second most important feature, highlighting its important role in the modeling process (Fig. 7(c–g)). Overall, the COD-in is reasonable for the most important feature for predicting the COD removal of AnMBR as it can be served as an important indicator of reactor loading. In contrast, other features, such as ambient temperature, influent temperature, and pH, exhibited lower importance in the model, suggesting that these environmental parameters have a limited impact on the AnMBR performance.

According to Fig. 7(b–d, f, h), significant differences exist in the feature importance rankings among the interpretation methods. For instance, the results of the model trained with additional data showed notable discrepancies for T-env, flux, and pH-in across the different interpretation methods. Because determining the optimal interpretation method is difficult, we recommend employing a ranking score that integrates the results of multiple methods to ensure robustness. In addition, we examined the changes in feature importance when additional data were included. The results indicated substantial shifts in feature importance rankings, deviating significantly from the results of model trained with additional data. The interpretation results of the model trained with the additional data (Fig. 7 (h)) were not identical to those of the original model (Fig. 7 (f)). The methods selected for interpretation were based on the structure and prediction of the models. Thus, the deviations between them can be attributed to the incorporation of additional data that may introduce noise into the model, potentially reducing the predictive performance. This result supports the findings in Section 3.3 indicating that the additional data used in this study did not contribute to developing a robust AnMBR ML model.

The variability of feature importance reflects the uncertainty associated with different mathematical methods for calculating the feature importance. However, despite these fluctuations, the top-tier features (COD-in and/or OD) consistently ranked highest across methods and models. This consistently suggests that while the lower-ranked variables carry uncertainty, the identification of the primary features is robust. The dominance of COD-in highlights the sensitivity to COD loadings, suggesting that COD-in can be a warning signal for COD-re prediction. For instance, when a high COD load is detected, the system can automatically reduce the inflow rate to give the biological process more time to treat the water. The importance of OD underscores the non-stationary nature of the AnMBR process (e.g., membrane fouling and biomass maturation). This implies that process monitoring should not rely solely on instantaneous sensor readings but must also incorporate time-dependent maintenance schedules.

3.5. Limitations and future perspectives

While this study indicates AutoML as a potential tool for AnMBR modeling with limited data, several limitations warrant attention. First, although AutoML outperformed previous models under the identical fixed data split strategy, these results should be interpreted as indicative within this specific experimental setup rather than definitively superior across all models and scenarios. Second, the observation that certain features (e.g., MLSS and MLVSS) contributed less to the model should not be interpreted as the physical or biological aspect, but reflects modeling results. Thus, these findings represent statistical feature importance, not mechanistic causality. Third, the performance failed to improve with a larger dataset does not imply data volume being useless. Instead, it underscores the critical importance of data consistency and quality over mere quantity, highlighting the impact of distribution shifts in dynamic processes of AnMBR treating wastewater. Finally, this study provides deterministic point estimates. To further enhance operational decision-making, future works could integrate probabilistic approaches. Such approaches can provide prediction intervals and quantify

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Lai, Y., Xiao, K., He, Y., Liu, X., Tan, J., Xue, W., Zhang, A., Huang, X., 2024. Machine learning for membrane bioreactor research: principles, methods, applications, and a tutorial. *Front. Environ. Sci. Eng.* 19, 34. <https://doi.org/10.1007/s11783-025-1954-2>.

Li, Y., Wang, Z., Xie, Y., Ding, B., Zeng, K., Zhang, C., 2021. AutoML: from methodology to application. *Trong: Proceedings of the 30th ACM International Conference on Information & Knowledge Management, Cikm '21*. Association for Computing Machinery, New York, NY, USA, pp. 4853–4856. <https://doi.org/10.1145/3459637.3483279>.

Li, Z., Jiang, T., Zhao, P., Qu, D., Liu, X., 2024. Enhanced DCH4 recovery via wave pulse mixing coupled in-situ hydrophobic membrane in AnMBR treating municipal wastewater. *Resour. Conserv. Recycl.* 209, 107773. <https://doi.org/10.1016/j.resconrec.2024.107773>.

Liu, Y., Yang, Z., Zou, X., Ma, S., Liu, D., Avdeev, M., Shi, S., 2023. Data quantity governance for machine learning in materials science. *Natl. Sci. Rev.* 10. <https://doi.org/10.1093/nsr/nwad125>.

Lundberg, S.M., Nair, B., Vavilala, M.S., Horibe, M., Eisses, M.J., Adams, T., Liston, D.E., Low, D.K.-W., Newman, S.-F., Kim, J., Lee, S.-I., 2018. Explainable machine-learning predictions for the prevention of hypoxaemia during surgery. *Nat. Biomed. Eng.* 2, 749–760. <https://doi.org/10.1038/s41551-018-0304-0>.

Mahanna, H., El-Rashidy, N., Kaloop, M.R., El-Sapakh, S., Alluqmani, A., Hassan, R., 2024. Prediction of wastewater treatment plant performance through machine learning techniques. *Desalination Water Treat.* 319, 100524. <https://doi.org/10.1016/j.dwt.2024.100524>.

uncertainty, offering more comprehensive information for biological wastewater treatment modeling.

4. Conclusions

Modeling wastewater treatment processes deepens mechanisms insight and reduces experimental resources. However, practitioners without data-driven expertise often struggle with model selection and tuning, limiting the practical application of machine learning. To address this challenge, this study applied AutoML to AnMBR modeling in wastewater treatment. The results demonstrated that AutoML improved modeling performance compared to the previous study. An in-depth investigation of input features and data volume revealed that the features selected through modeling significantly improved the capability of ML to model the AnMBR process. By incorporating reactor OD, the R² value further increased. However, increasing data volume did not consistently enhance model performance, underscoring the importance of data quality in wastewater treatment modeling. Based on these findings, using AnMBR modeling as a case study, this work demonstrates the efficiency and convenience of AutoML in small-sample modeling and ablation analysis. Furthermore, an integrated feature importance analysis method was proposed to evaluate the feature importance for AutoML model applications. Overall, this study provides new perspectives and references for advancing the application of AutoML in wastewater treatment, especially for modeling in data-scarce scenarios such as laboratory-scale and pilot-scale experiments.

CRedit authorship contribution statement

Junjie Yu: Writing – original draft, Visualization, Software, Investigation. **Zhonghua Zheng:** Writing – review & editing, Resources, Methodology, Investigation. **Jialing Ni:** Writing – review & editing, Resources. **Jiayuan Ji:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2026.128801>.

Data availability

The authors do not have permission to share data.

References

Breiman, L., 2001. Random forests. *Mach. Learn.* 45, 5–32. <https://doi.org/10.1023/A:1010933404324>.
Bridgewater, L.L., Baird, R.B., Eaton, A.D., Rice, E.W., 2017. American public health association, American water works association, water environment Federation.

Standard Methods for the Examination of Water and Wastewater, 23rd edition. American Public Health Association, Washington, DC.
Chen, T., Guestrin, C., 2016. XGBoost: a scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '16. Association for Computing Machinery, New York, NY, USA, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>.
Dong, W., Yuan, H., Lin, W., Liu, Z., Xiang, J., Wei, Z., Li, L., Li, Q., Dai, Y., 2025. A global urban tree leaf area index dataset for urban climate modeling. *Sci. Data* 12, 426. <https://doi.org/10.1038/s41597-025-04729-y>.
Dorogush, A.V., Ershov, V., Gulin, A., 2018. CatBoost: gradient boosting with categorical features support. <https://doi.org/10.48550/arXiv.1810.11363>.
Fan, F.J., Shi, Y., 2022. Effects of data quality and quantity on deep learning for protein-ligand binding affinity prediction. *Bioorg. Med. Chem.* 72, 117003. <https://doi.org/10.1016/j.bmc.2022.117003>.
Faraway, J.J., Augustin, N.H., 2018. When small data beats big data. *Role Stat. Era Big Data* 136, 142–145. <https://doi.org/10.1016/j.spl.2018.02.031>.
Geurts, P., Ernst, D., Wehenkel, L., 2006. Extremely randomized trees. *Mach. Learn* 63, 3–42. <https://doi.org/10.1007/s10994-006-6226-1>.
Grinsztajn, L., Oyallon, E., Varoquaux, G., 2022. Why do tree-based models still outperform deep learning on typical tabular data? In: Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., Oh, A. (Eds.), *Advances in Neural Information Processing Systems*. Curran Associates, Inc., pp. 507–520.
He, X., Zhao, K., Chu, X., 2021. AutoML: a survey of the state-of-the-art. *Knowl.-Based Syst.* 212, 106622. <https://doi.org/10.1016/j.knosys.2020.106622>.
Ho, J., Sung, S., 2010. Methanogenic activities in anaerobic membrane bioreactors (AnMBR) treating synthetic municipal wastewater. *Bioresour. Technol.* 101, 2191–2196. <https://doi.org/10.1016/j.biortech.2009.11.042>.
Hu, D., Su, H., Chen, Z., Cui, Y., Ran, C., Xu, J., Xiao, T., Li, X., Wang, H., Tian, Y., Ren, N., 2017. Performance evaluation and microbial community dynamics in a novel AnMBR for treating antibiotic solvent wastewater. *Bioresour. Technol.* 243, 218–227. <https://doi.org/10.1016/j.biortech.2017.06.095>.
Ji, J., Chen, Y., Hu, Y., Ohtsu, A., Ni, J., Li, Y., Sakuma, S., Hojo, T., Chen, R., Li, Y.-Y., 2021a. One-year operation of a 20-L submerged anaerobic membrane bioreactor for real domestic wastewater treatment at room temperature: pursuing the optimal HRT and sustainable flux. *Sci. Total Environ.* 775, 145799. <https://doi.org/10.1016/j.scitotenv.2021.145799>.
Ji, J., Ni, J., Ohtsu, A., Isozumi, N., Hu, Y., Du, R., Chen, Y., Qin, Y., Kubota, K., Li, Y.-Y., 2021b. Important effects of temperature on treating real municipal wastewater by a submerged anaerobic membrane bioreactor: removal efficiency, biogas, and microbial community. *Bioresour. Technol.* 336, 125306. <https://doi.org/10.1016/j.biortech.2021.125306>.
Ji, J., Sakuma, S., Ni, J., Chen, Y., Hu, Y., Ohtsu, A., Chen, R., Cheng, H., Qin, Y., Hojo, T., Kubota, K., Li, Y.-Y., 2020. Application of two anaerobic membrane bioreactors with different pore size membranes for municipal wastewater treatment. *Sci. Total Environ.* 745, 140903. <https://doi.org/10.1016/j.scitotenv.2020.140903>.
Jiang, F., Zheng, Z., Coe, H., Healy, R.M., Poulain, L., Gros, V., Zhang, H., Li, W., Liu, D., West, M., Topping, D., Riemer, N., 2025. Integrating simulations and observations: a foundation model for estimating the aerosol mixing state index. *ACS EST Air* 2, 877–890. <https://doi.org/10.1021/acsestair.4c00329>.
Jury, W.A., Vaux, H.J., 2007. The emerging global water crisis: managing scarcity and conflict between water users. In: *Advances in Agronomy*. Elsevier, pp. 1–76. [https://doi.org/10.1016/S0065-2113\(07\)95001-4](https://doi.org/10.1016/S0065-2113(07)95001-4).
Karmaker (“Santu”), S.K., Hassan, Md.M., Smith, M.J., Xu, L., Zhai, C., Veeramachaneni, K., 2021. AutoML to date and beyond: challenges and opportunities. *ACM Comput. Surv.* 54. <https://doi.org/10.1145/3470918>.
Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y., 2017. LightGBM: a highly efficient gradient boosting decision tree. In: *Advances in Neural Information Processing Systems*. Curran Associates, Inc.
Kokol, P., Kokol, M., Zagoranski, S., 2022. Machine learning on small size samples: a synthetic knowledge synthesis. *Sci. Prog.* 105, 00368504211029777. <https://doi.org/10.1177/00368504211029777>.
Krzeminski, P., Leverette, L., Malamis, S., Katsou, E., 2017. Membrane bioreactors – a review on recent developments in energy reduction, fouling control, novel configurations, LCA and market prospects. *J. Membr. Sci.* 527, 207–227. <https://doi.org/10.1016/j.memsci.2016.12.010>.
Lai, Y., Xiao, K., He, Y., Liu, X., Tan, J., Xue, W., Zhang, A., Huang, X., 2024. Machine learning for membrane bioreactor research: principles, methods, applications, and a tutorial. *Front. Environ. Sci. Eng.* 19, 34. <https://doi.org/10.1007/s11783-025-1954-2>.
Li, G., Ji, J., Ni, J., Wang, S., Guo, Y., Hu, Y., Liu, S., Huang, S.-F., Li, Y.-Y., 2022. Application of deep learning for predicting the treatment performance of real municipal wastewater based on one-year operation of two anaerobic membrane bioreactors. *Sci. Total Environ.* 813, 151920. <https://doi.org/10.1016/j.scitotenv.2021.151920>.
Li, Q., Zhao, W., Cui, S., Gadow, S.I., Qin, Y., Li, Y.-Y., 2024. Synergetic association of hydroxyapatite-mediated biofilm and suspended sludge enhances resilience of partial nitrification/anammox (PN/A) system treating high-strength anaerobic membrane bioreactor (AnMBR) permeate. *Bioresour. Technol.* 412, 131391. <https://doi.org/10.1016/j.biortech.2024.131391>.
Li, Y., Wang, Z., Xie, Y., Ding, B., Zeng, K., Zhang, C., 2021. AutoML: from methodology to application. In: Proceedings of the 30th ACM International Conference on Information & Knowledge Management, Cikm '21. Association for Computing Machinery, New York, NY, USA, pp. 4853–4856. <https://doi.org/10.1145/3459637.3483279>.
Li, Z., Jiang, T., Zhao, P., Qu, D., Liu, X., 2024. Enhanced DCH4 recovery via wave pulse mixing coupled in-situ hydrophobic membrane in AnMBR treating municipal

van Loosdrecht, M.C.M., Brdjanovic, D., 2014. Anticipating the next century of wastewater treatment. *Science* 344, 1452–1453. <https://doi.org/10.1126/science.1255183>.

Moideen, S.N.F., Krishnan, S., Li, Y.-Y., Hassim, M.H., Kamyab, H., Nasrullah, M., Din, M. F.M., Halim, K.A., Chaiprapat, S., 2023. Performance evaluation and energy potential analysis of anaerobic membrane bioreactor (AnMBR) in the treatment of simulated milk wastewater. *Chemosphere* 317, 137923. <https://doi.org/10.1016/j.chemosphere.2023.137923>.

Pretel, R., Dur’an, F., Robles, A., Ruano, M.V., Ribes, J., Serralta, J., Ferrer, J., 2015. Designing an AnMBR-based WWTP for energy recovery from urban wastewater: the role of primary settling and anaerobic digestion. *Sep. Purif. Technol.* 156, 132–139. <https://doi.org/10.1016/j.seppur.2015.09.047>.

Robles, ‘A., Jim’enez-Benítez, A., Gim’enez, J.B., Dur’an, F., Ribes, J., Serralta, J., Ferrer, J., Rogalla, F., Seco, A., 2022. A semi-industrial scale AnMBR for municipal wastewater treatment at ambient temperature: performance of the biological process. *Water Res.* 215, 118249. <https://doi.org/10.1016/j.watres.2022.118249>.

Robles, ‘A., Ruano, M.V., Charfi, A., Lesage, G., Heran, M., Harmand, J., Seco, A., Steyer, J.-P., Batstone, D.J., Kim, J., Ferrer, J., 2018. A review on anaerobic membrane bioreactors (AnMBRs) focused on modelling and control aspects. *Bioresour. Technol.* 270, 612–626. <https://doi.org/10.1016/j.biortech.2018.09.049>.

Shwartz-Ziv, R., Armon, A., 2022. Tabular data: deep learning is not all you need. *Inf. Fusion* 81, 84–90. <https://doi.org/10.1016/j.inffus.2021.11.011>.

Uman, A.E., Bair, R.A., Yeh, D.H., 2021. Assessment of an anaerobic membrane bioreactor (AnMBR) treating medium-strength synthetic wastewater under cyclical membrane operation. *Membranes* 11. <https://doi.org/10.3390/membranes11060415>.

- wastewater. *Resour. Conserv. Recycl.* 209, 107773. <https://doi.org/10.1016/j.resconrec.2024.107773>.
- Liu, Y., Yang, Z., Zou, X., Ma, S., Liu, D., Avdeev, M., Shi, S., 2023. Data quantity governance for machine learning in materials science. *Natl. Sci. Rev.* 10. <https://doi.org/10.1093/nsr/nwad125> nwad125.
- Lundberg, S.M., Lee, S.-I., 2017. A unified approach to interpreting model predictions. In: *Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS'17*. Curran Associates Inc., Red Hook, NY, USA, pp. 4768–4777.
- Lundberg, S.M., Nair, B., Vavilala, M.S., Horibe, M., Eisses, M.J., Adams, T., Liston, D.E., Low, D.K.-W., Newman, S.-F., Kim, J., Lee, S.-I., 2018. Explainable machine-learning predictions for the prevention of hypoxaemia during surgery. *Nat. Biomed. Eng.* 2, 749–760. <https://doi.org/10.1038/s41551-018-0304-0>.
- Mahanna, H., El-Rashidy, N., Kaloop, M.R., El-Sapakh, S., Alluqmani, A., Hassan, R., 2024. Prediction of wastewater treatment plant performance through machine learning techniques. *Desalination Water Treat.* 319, 100524. <https://doi.org/10.1016/j.dwt.2024.100524>.
- Martin Bland, J., Altman, DouglasG., 1986. Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet* 327, 307–310. [https://doi.org/10.1016/S0140-6736\(86\)90837-8](https://doi.org/10.1016/S0140-6736(86)90837-8).
- McElfresh, D., Khandagale, S., Valverde, J., Prasad C, V., Ramakrishnan, G., Goldblum, M., White, C., 2023. When do neural nets outperform boosted trees on tabular data? In: Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., Levine, S. (Eds.), *Advances in Neural Information Processing Systems. Curran Associates, Inc.*, pp. 76336–76369.
- Moideen, S.N.F., Krishnan, S., Li, Y.-Y., Hassim, M.H., Kamyab, H., Nasrullah, M., Din, M. F.M., Halim, K.A., Chaiprapat, S., 2023. Performance evaluation and energy potential analysis of anaerobic membrane bioreactor (AnMBR) in the treatment of simulated milk wastewater. *Chemosphere* 317, 137923. <https://doi.org/10.1016/j.chemosphere.2023.137923>.
- Pal, R., Arcamo, L., Farnood, R., 2024. Predicting the occurrence of substituted and unsubstituted, polycyclic aromatic compounds in coking wastewater treatment plant effluent using machine learning regression. *Chemosphere* 361, 142476. <https://doi.org/10.1016/j.chemosphere.2024.142476>.
- Pretel, R., Durán, F., Robles, A., Ruano, M.V., Ribes, J., Serralta, J., Ferrer, J., 2015. Designing an AnMBR-based WWTP for energy recovery from urban wastewater: the role of primary settling and anaerobic digestion. *Sep. Purif. Technol.* 156, 132–139. <https://doi.org/10.1016/j.seppur.2015.09.047>.
- Ren, L., Chen, X., Wu, J., Huang, S., Williams, A., Su, Q., 2024. Study on membrane fouling mechanisms and mitigation strategies in a pilot-scale anaerobic membrane bioreactor (P-AnMBR) treating digestate. *Water Res.* 263, 122166. <https://doi.org/10.1016/j.watres.2024.122166>.
- Robles, Á., Jiménez-Benítez, A., Giménez, J.B., Durán, F., Ribes, J., Serralta, J., Ferrer, J., Rogalla, F., Seco, A., 2022. A semi-industrial scale AnMBR for municipal wastewater treatment at ambient temperature: performance of the biological process. *Water Res.* 215, 118249. <https://doi.org/10.1016/j.watres.2022.118249>.
- Robles, Á., Ruano, M.V., Charfi, A., Lesage, G., Heran, M., Harmand, J., Seco, A., Steyer, J.-P., Batstone, D.J., Kim, J., Ferrer, J., 2018. A review on anaerobic membrane bioreactors (AnMBRs) focused on modelling and control aspects. *Bioresour. Technol.* 270, 612–626. <https://doi.org/10.1016/j.biortech.2018.09.049>.
- Schreck, J.S., Petzke, W., Jiménez, P.A., Brummet, T., Knievel, J.C., James, E., Kosović, B., Gagne, D.J., 2023. Machine learning and VIIRS satellite retrievals for skillful fuel moisture content monitoring in wildfire management. *Remote Sens.* 15, 3372. <https://doi.org/10.3390/rs15133372>.
- Shwartz-Ziv, R., Armon, A., 2022. Tabular data: deep learning is not all you need. *Inf. Fusion* 81, 84–90. <https://doi.org/10.1016/j.inffus.2021.11.011>.
- Truong, A., Walters, A., Goodsitt, J., Hines, K., Bruss, C.B., Farivar, R., 2019. Towards automated machine learning: evaluation and comparison of AutoML approaches and tools. In: 2019 IEEE 31st International Conference on Tools with Artificial Intelligence (ICTAI), pp. 1471–1479. <https://doi.org/10.1109/ICTAI.2019.00209>.
- Uman, A.E., Bair, R.A., Yeh, D.H., 2021. Assessment of an anaerobic membrane bioreactor (AnMBR) treating medium-strength synthetic wastewater under cyclical membrane operation. *Membranes* 11. <https://doi.org/10.3390/membranes11060415>.
- van Loosdrecht, M.C.M., Brdjanovic, D., 2014. Anticipating the next century of wastewater treatment. *Science* 344, 1452–1453. <https://doi.org/10.1126/science.1255183>.
- Wang, C., Wu, Q., Huang, S., Saied, A., 2021a. Economical hyperparameter optimization with blended search strategy. In: *The Ninth International Conference on Learning Representations (ICLR 2021)*.
- Wang, C., Wu, Q., Weimer, M., Zhu, E.Eric, 2021b. FLAML: a fast and lightweight AutoML library. Presented at the Fourth Conference on Machine Learning and Systems (MLSys 2021).
- Wang, T., Li, Y.-Y., 2024. Predictive modeling based on artificial neural networks for membrane fouling in a large pilot-scale anaerobic membrane bioreactor for treating real municipal wastewater. *Sci. Total Environ.* 912, 169164. <https://doi.org/10.1016/j.scitotenv.2023.169164>.
- Winkler, M.-K.H., van Loosdrecht, M.C.M., 2022. Intensifying existing urban wastewater. *Science* 375, 377–378. <https://doi.org/10.1126/science.abm3900>.
- Wu, Q., Wang, C., Huang, S., 2021. Frugal optimization for cost-related hyperparameters. *Proc. Thirty-Fifth AAAI Conf. Artif. Intell. AAAI-21*. <https://doi.org/10.48550/arXiv.2005.01571>.
- Xia, Xinran, Fu, D., Shao, W., Jiang, R., Wu, S., Zhang, P., Yang, D., Xia, Xiangao, 2023. Retrieving precipitable water vapor over land from satellite passive microwave radiometer measurements using automated machine learning. *Geophys. Res. Lett.* 50. <https://doi.org/10.1029/2023GL105197> e2023GL105197.
- Xie, Z., Wang, Z., Wang, Q., Zhu, C., Wu, Z., 2014. An anaerobic dynamic membrane bioreactor (AnDMBR) for landfill leachate treatment: performance and microbial community identification. *Bioresour. Technol.* 161, 29–39. <https://doi.org/10.1016/j.biortech.2014.03.014>.
- Yu, J., Zheng, Z., Lindley, S., Zhao, L., Wang, C., Wu, Q., Li, L., Topping, D.O., Schreck, J. S., Gagne, D.J., Oleson, K.W., 2025. Leveraging automated machine learning (AutoML) for urban climate emulation. *Big Data Earth Syst I*, 100040. <https://doi.org/10.1016/j.bdes.2025.100040>.
- Zheng, Z., Fiore, A.M., Westervelt, D.M., Milly, G.P., Goldsmith, J., Karambelas, A., Curci, G., Randles, C.A., Paiva, A.R., Wang, C., Wu, Q., Dey, S., 2023. Automated machine learning to evaluate the information content of tropospheric trace gas columns for fine particle estimates over India: a modeling testbed. *J. Adv. Model. Earth Syst.* 14, 17727–17741. <https://doi.org/10.1029/2022ms003099>.
- Wang, T., Li, Y.-Y., 2024. Predictive modeling based on artificial neural networks for membrane fouling in a large pilot-scale anaerobic membrane bioreactor for treating real municipal wastewater. *Sci. Total Environ.* 912, 169164. <https://doi.org/10.1016/j.scitotenv.2023.169164>.